

## DIVERSITY, EQUITY, AND INCLUSION

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Western mathematics has an unfortunate history of being intimidating and exclusionary. One major effect of this is that several minority groups are now underrepresented in mathematics. To make progress towards a more representative and equitable environment, both math institutions and individuals must do their part to combat racism, homophobia, transphobia, and other prejudices. The following is a description of actions that I have taken to make mathematics a more inclusive pursuit, as well as my future plans to help underrepresented minorities.

The student body at my graduate institution (UIC) is comprised of over 60% students of color, and it is primarily sourced from the urban Chicago Public Schools system. UIC is also both a Hispanic-Serving Institution and an Asian American and Native American Pacific Islander-Serving Institution. In addition, many students are first generation university students. My teaching methods partially developed over my six years at UIC in response to the diverse needs of the various learning styles of these undergraduates. These methods were further refined by my time at Boston College and at Wellesley, a historically women's college. I frequently employ active learning strategies in my classrooms, since active learning has the potential to increase the success of those typically underrepresented in mathematics [LHKW14],[EH14]. Active learning classes take the diversity of students into account not just by giving a voice to those who may have been ignored before, but also by soliciting more perspectives when problem-solving. Furthermore, I recognize that not all students have the same background and that I may need to offer more time and resources to certain students in order to make my classroom an equitable one.

During my time at Boston College and continuing at Wellesley, I have tried to make my courses as accessible and welcoming as possible. One way I do this is by making my classes cost-free. By this I mean I do not make my students spend money on course material. Textbooks are suggested but not required. My lectures are self contained and in some cases (Calculus 1 and 2) I have gone as far as to prepare worksheets/packets on course topics for my students to reference. I also use free and open source software, examples being Desmos, WebWork, and SageMath, so that students do not have to purchase subscriptions to alternatives. These choices make my classes more accessible to students from lower socio-economic backgrounds, which have a correlation with some underrepresented groups. I also have used class discussion boards to help connect my students with each other to build a classroom community. Towards connecting students to the broader community, I frequently have a homework question that has them review websites devoted to underrepresented groups in math (e.g., [wom25], [Lat25], [MGB25], [ind25]) and find someone they identify with, explaining to me what is inspiring about that person. This helps them identify with another person in mathematics and realize there is a place for them too.

For a time during graduate school, I spent time as a tutor for the Asian American Resource and Cultural Center where I became more familiar with the challenges facing minority immigrant students. It was a valuable experience leaning how the ways questions are written can confuse those for whom English is a second language. Later on, over my last 3 years at UIC, I had the pleasure of mentoring a hispanic undergraduate in mathematics. They started as a student in my multivariable calculus workshop, and every semester after we would meet once a week to discuss math and their life in general. These meetings helped me gain an understanding of how challenging the current math environment is for these students. For example, during one meeting they explained to me

how for their whole life they had been told by others that they were bad at math, and should focus on other topics. After graduation, they went on to obtain a master degree in mathematics. I would like to mentor more students in the future, both formally and informally.

As a gay graduate student, I felt there was not a strong LGBT community in my department. In response to this, another queer student and I started the Out in Math seminar at UIC. This was a monthly meeting for LGBT mathematics graduate students and allies during which we would discuss issues related to the LGBT community and mathematics. This also helped to raise LGBT visibility in our department. At several past Joint Mathematics Meetings, I attended an event hosted by Spectra, the professional association for LGBT Mathematicians [Spe]. After some discussion, and with their support, I converted the Out in Math seminar into the first student chapter of Spectra.

More recently, I was an organizer of the Boston College satellite conference of Math For All [Mat24]. The goal of this conference is to provide a welcoming and inclusive environment for participants that may typically forgo more standard conferences. We also have a low barrier to entry so that participants can get the valuable exposure and networking experience that those from underrepresented groups usually have missed. This event has been valuable to attendees and some have attributed the internships they obtained to participating in Math For All.

I intend to continue these activities during the next steps in my career. I hope to start more seminars and organize more conferences aimed at increasing community for minority mathematicians. For example, another goal of mine is to start an LGBT invited speaker session to expose graduate students to modern queer mathematicians. In 2019 I attended at the University of Michigan the LG&TBQ conference for queer geometers and topologists [LG&19], and I was inspired by its success. I hope to be involved with organizing similar events in the future.

## REFERENCES

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